

Biverbal predicate constructions and related concepts: Serial verbs, converbs, auxiliaries, and grammaticalization

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Quite a few of the technical terms used in research on “complex predicates” are understood differently by different researchers, or have a vague or unclear meaning. This paper offers discussions and definitions of the terms *serial verb construction*, *converb*, and *auxiliary*, as well as the terms *grammaticalization* and *inventorization* (corresponding to “lexicalization”). The definitions rely on other well-defined terms and are largely in line with earlier usage, but of course not entirely. I also propose the novel term *biverbal predicate construction* that is defined in a simple way and does not rely on the vague term “complex predicate”. The paper provides some general principles of good comparative-concept terms and reflects on the reasons for terminological divergence in linguistics.

1. Three retro-definitions

This paper discusses a range of concepts and technical terms that are important for complex predicate research and proposes a number of retro-definitions for terms that have been widely used in the past. The definitions that I propose are not always the same as the definitions used elsewhere in this volume, and the paper will discuss some of these discrepancies. I will also point out the difference between *descriptive categories* used for particular languages on the one hand, and *comparative-concept terms* on the other. I will discuss three types of definitions for comparative terms: retro-definitions, novel-term definitions, and ad hoc definitions. In order to contrast this conceptually-oriented work with empirical work, I will briefly discuss two empirical hypotheses that can be assessed on the basis of some of the definitions (in §8), but these will not be a major focus of the paper.

In order to start as concretely as possible, I begin with the definitions of three well-known terms that play a role for the domain that is known by the umbrella term “complex predicates”. As these three labels are widely used in the earlier literature, (1)-(3) are retro-definitions, a notion that is explained in §2.3 below. They will be discussed and exemplified extensively in later sections of this paper.

(1) **serial verb construction** (§4)

A serial verb construction is a monoclausal construction consisting of multiple independent verbs with no element linking them and with no semantic predicate–argument relation between the verbs. (Haspelmath 2016: 296)

(2) **converb** (§5)

A converb is a form that (i) consists of a verb stem and a special affix but no subject person indexing, (ii) that cannot occur as the predicate of an independent sentence, and (iii) that can occur as the predicate of a subordinate clause that is neither a relative clause nor a complement clause.

(3) **auxiliary** (§6)

An auxiliary is a non-affixed bound form cooccurring with a verb that expresses TAMEP (tense, aspect, modality, evidentiality, polarity) meanings and that has person marking which indexes the verb’s subject.

The proposed definitions of *serial verb construction* in (1) and *converb* in (2) are probably of substantial practical utility for general linguistics, whereas I would think that the definition of *auxiliary* is less useful. The purpose of this paper is to advance our methodology by contributing to awareness about problems that arise with these terms and their definitions, as well as to awareness about the limitations of comparative-concept terms.

I do not suggest a definition of the term “complex predicate construction” here, as it has been used in very heterogeneous ways in the literature (see §3), but I will discuss and define a related novel term: *biverbal predicate construction*, defined as in (4).

(4) biverbal predicate construction (§3)

A biverbal predicate construction is a monoclausal construction with a predicate that includes two verb roots.

Before discussing the three retro-defined terms and the novel term, I will discuss the nature of comparative concepts in (§2), and later in the paper, I will discuss the terms “grammaticalization” and “lexicalization”, which are important in complex predicate research (§7). For the diachronic domain, too, I think that rigorous conceptual analysis can help us in framing the issues in productive ways.

2. Four requirements on comparative concept terms

Comparative concepts are the basis of comparative and general linguistics, and many technical terms are routinely used in a general sense by linguists. This paper discusses four requirements for comparative concept terms that are not always kept in mind by comparative linguists. They are summarized in (4a-d).

- (4) a. Comparative terms must be applied uniformly, i.e. using the same criteria in all languages (§2.1).
- b. Traditional labels should be defined in such a way that they largely conform to traditional usage (§2.2).
- c. Commonly used traditional terms should have general definitions, i.e. we need retro-definitions for them (§2.3).
- d. Comparative terms should be useful for formulating generalizations (§2.4).

My focus here will be on (4a-c), because even though many authors emphasize (4d) (the usefulness of comparative concepts), I do not actually think that it is as important as is often thought (contrary to what I said in Haspelmath 2010: 670). Throughout this paper, it should be remembered that comparative concepts are methodological tools, not theoretical hypotheses. Thus, we are talking about methodological requirements for comparative research, not about general-theoretical claims. (And we are not talking about language-particular analyses, a topic that many of the papers in this volume are concerned with.)

2.1. Uniform application

Comparative concepts must be identified in the same way across all languages. This is similar in other kinds of cross-cultural research, e.g. in comparative legal or political studies (where notions such as “children’s rights” and “democracy” must be defined in the same way for all countries). In linguistics, there is a long tradition of carrying over

terminology from one language to another in an unconstrained way, so that our grammatical terms often have different meanings in different contexts and cannot be used easily for comparative purposes.

For example, terms such as “monoclausal” or “phonological word” may be understood in fairly different ways, using different criteria in different languages (e.g. Butt 2010: 57; Aikhenvald et al. 2020: §2). But for a comparative project, one needs a uniformly applicable definition. For “monoclausal”, I propose the definition in (5) (see Haspelmath 2016: §2.2; and some further discussion in §3 below).

(5) **monoclausal construction**

A monoclausal construction is a clausal construction that can have only one predicate negation, even if there are multiple predicate forms.

For the term “phonological word”, by contrast, I do not know of any work where a uniformly applicable definition is proposed. In Haspelmath (2023b), I give a definition of “(grammatical) word”, and I think it is probably best if “word” is only used in a grammatical sense. Phonological domains can be very diverse, and the extent to which they correlate with grammatical domains is an empirical question that does not have a clear answer at the moment (see Bickel et al. 2009; but there is little follow-up work in this tradition). It has sometimes been proposed that certain kinds of “lexical” phonological alternations occur only in certain kinds of “lexical” grammatical domains (e.g. Mohanan 1986; Rubach 2008), but the phenomena of the world’s languages do not always bear this out. The old claims may still be true as a tendency, but they have not been tested systematically.¹

Thus, there is no general definition of “phonological word”, and the same applies to some other terms that are frequently mentioned in the context of “complex predicates”, e.g. “phrase”, “head/dependent”, “nucleus/core”, “(non)finite”, “adverb(ial)”. I will avoid using these terms, even though one may eventually want to retro-define them (as per 4c). It should also be noted that the definitions must be strictly synchronic. While diachronic information is available for many well-known languages, this cannot be relevant for comparative concepts.

2.2. Traditional labels and traditional usage

Although many linguists are aware of widespread terminological problems in grammatical research, traditional terms are typically treated as if they have a single meaning that largely corresponds to traditional usage. For this reason, traditional terms should not be used in drastically novel ways.

If one wanted to define the term *complex predicate*, it would thus be confusing to propose a definition that does not correspond to the traditional usage. The term is fairly recent (going back to the 1980s, as noted below in §3) and has already been used in a range of different ways, but depictive and manner expressions have not normally been included. Thus, it is surprising that Croft (2022: 432) includes them in his notion of “complex predicate”, along with resultative complex predicates (which had been treated as such in the earlier literature):

¹ Vowel alternations such as the *e/a* alternation reported for Circassian by Arkadiev (in this volume) are surely more common in affixal or compound-type constructions than in phrase-type constructions, but vowel harmony is attested in clitic constructions (e.g. in Turkish), so these patterns can hardly be taken as criterial for “morphological” status. (See also Haspelmath 2023c on clitics and the problems with the notion of “phonological attachment”).

- (6) “resultative complex predicate”: *We painted the door red.*
 “depictive complex predicate”: *I ate the carrots raw.*
 “manner complex predicate”: *We crawled down the slope slowly.*

The expressions *eat raw* and *crawl slowly* exhibit similarities with *paint red*, but they have not been treated as “complex predicates” in the literature before, so Croft’s novel usage is potentially confusing.

It is rarely possible to define a term in such a way that the definition corresponds fully to earlier general usage, but the principle should be that it corresponds very largely to its earlier implicit understanding. If a term for a relatively new concept is needed, a novel term should be coined (like the term *biverbal predicate construction* in (4)).

2.3. Commonly used terms should be retro-defined

For the three terms that we saw in §1, the problem is not that they have been used in non-traditional ways. They are used in similar ways across the literature, but they are rarely defined in a rigorous way. Thus, the definitions proposed here may be important contributions, even though they are not based on empirical findings.

Now one might object that everyday terms like “auxiliary” or “clitic” or “affix” can never be defined in a strict way, because the reality that they are meant to describe is highly complex and multidimensional, with gradients rather than sharp boundaries. However, in order to show that there is a gradient or that there are multiple dimensions, one needs sharply defined concepts. Perhaps one should coin novel terms for such sharply defined concepts, but as linguists will continue to use the old terms, these also need to be defined clearly. I am not suggesting that the old terms can necessarily lead to deeper understanding (that they are necessarily useful, cf. (4d)), but as they are widely applied to cross-linguistic and general-theoretical research, we should not treat them as undefined. If they were used without a definition, they could be understood only as stereotypes, and this would not lead to progress in understanding.

2.4. Comparative concepts need not be used for generalizations

In my earlier work, I emphasized that while descriptive (or analytical) concepts allow us to describe languages, comparative concepts allow us to compare languages with the goal of formulating universals. Thus, I listed nine universals of indexing in Haspelmath (2013), ten universals of serial verb constructions in Haspelmath (2016), and eight universals of reflexive constructions in Haspelmath (2023a).

However, I have come to realize that sharp definitions are useful even if we have no particular universal claims in mind. For example, a retro-definition of “affix” (Haspelmath 2021a) is useful because it allows us to (retro-)define further well-known terms such as “clitic”, “word” and “inflection”. All of these are commonly used, and if we have no definition of them, we cannot make precise statements and we risk perpetuating unsubstantiated stereotypes. Whether they are needed for generalizations is a separate question.

Moreover, not all comparisons have the goal of finding universal generalizations. We may want to compare language structures for all kinds of purposes, including psycholinguistic, sociolinguistic or historical studies. Thus, rather than saying that comparative terms are used only as the basis of universals, it suffices that they should be useful for testing general claims or simply for talking about salient differences between languages.

3. Verbal complex predicate constructions and other kinds of “complex predicates”

As noted by Andrews (2023), “the term COMPLEX PREDICATE has been widely and rather loosely applied to a variety of constructions where for some reason it appears that two predicates that might be regarded as independent are behaving as one”. There are a wide variety of phenomena that have been given this label, and I am not going to suggest a retro-definition here. It seems that “complex predicate” can only be understood as a vague “umbrella term” for a range of loosely related phenomena which have been discussed under this label. As Krauß & Harvey (2021: 102) note, the term has been much more often used than defined explicitly, and as a result, its uses are united by family resemblances rather than by a unified concept.

However, I propose the novel term *biverbal predicate construction*, defined in (7), and I briefly compare it with the term *verbal complex predicate construction* as defined by Bisang et al. (this volume), as in (8).

(7) **biverbal predicate construction**

A biverbal predicate construction is a monoclausal construction with a predicate that includes two verb roots.

(8) **verbal complex predicate construction**

A verbal complex predicate (VCP) is a monoclausal construction with a single set of argument positions, consisting of at least two verbs (or “verb-like” items), which either (i) both belong synchronically to the class of verbs, or (ii) combine a lexical verb with a grammatical element which can also be used as a full verb in other contexts.

Extensionally, these two definitions are (apparently) quite similar, but the definition of *biverbal predicate construction* is simpler and clearer. It does not make use of “a single set of argument positions”, and of the notions “verb-like item”, “lexical verb”, and “full verb”.² It mentions verb roots, but a “word” notion plays no role in it, unlike in some other definitions of complex predicates (e.g. Bower 2014: 266-267). The restriction to two verbs additionally makes the definition simpler, at little cost (more than two verbs occur quite rarely).

The condition that there is a single set of argument positions will be discussed further below in this section. The distinction between “lexical verbs” and “grammatical elements which can also be used as full verbs” is not made in the definition of biverbal predicate constructions in (7) because the definition presupposes that we distinguish between a polysemous linguistic form and a pair of homonyms. For example, when classifying the Spanish construction in (9a) (where *seguir* is used in an aspectual sense, not in the original sense ‘follow’), we need to determine whether *seguir* in (9a) is the same verb as in (9b).

(9) Spanish

- a. *Los niños seguían jugando.*
 the children continued playing
 ‘The children continued to play.’

² Another way in which it is clearer is that the term is compositional (a biverbal predicate construction is a kind of *predicate construction*, and a kind of *biverbal construction*), whereas “verbal complex predicate construction” is not defined as a kind of “complex predicate construction”. The latter is not clearly defined, as noted by Andrews (2023) and others.

- b. *Los niños seguían a sus padres.*
 the children followed ACC their parents
 ‘The children followed their parents.’

If we are dealing with the same (polysemous) verb in both cases, then (9a) is a biverbal construction, but if *seguían* in (9a) and (9b) are two homonymous forms (only diachronically related), then it is not a biverbal construction. Perhaps the condition (ii) of the definition in (8) is meant to include cases where the grammatical element and its lexemic source are only diachronically related, but then it would not be a strictly synchronic definition (and it would thus violate the condition of uniform application in (4a)).

Both biverbal predicate constructions and verbal complex predicate constructions are defined rather narrowly in (7)-(8), compared to the way the term “complex predicate” has been used and defined elsewhere in the literature. The term originates in the Stanford tradition of LFG and HPSG, in works such as Ishikawa (1985), Alsina (1993), Butt (1993), Alsina et al. (1997), Andrews & Manning (1999). According to Butt (1993: 2), a complex predicate construction is one in which two or more semantic heads contribute arguments but the grammatical functional structure (or valency) is that of a simple predicate. The definition thus makes crucial use of the level of “grammatical functions” in LFG, and it even extends to affixally coded voice constructions such as causatives and applicatives (e.g. in Alsina 1993; Lomashvili 2011). Other definitions are still broader, e.g. Bowerman (2010: 42):

“Complex predicates are structures in which more than one element in the clause contributes information to the predicate which is normally associated with a verbal head.”

If tense and mood information is “normally associated with a verbal head”,³ then all constructions with tense and mood auxiliaries count as complex predicates (see also Godard & Samvelian (2021: 419-420), who discuss “complex predicates” in an HPSG context and also include all kinds of auxiliaries).

Let me now say a few more things about the four crucial terms included in the definition of biverbal predicate construction (*verb*, *root*, *monoclausal*, and *predicate*), as well as about the notion of a “single set of argument positions” that is not part of it but occurs in the definition of *verbal complex predicate* in (8).

In a comparative context, a *verb* is best defined as a morph (i.e. a minimal form) that denotes an action (Haspelmath 2023d: 34), i.e. it is defined in substantive terms rather than with respect to formal properties that can vary across languages. A *root* is a minimal form with contentful meaning, by which grammatical meanings such as tense or aspect are excluded. Moreover, a root must be able to occur on its own without another root, which excludes causative (and other action-denoting) affixes. The criterion of denoting an “action” is admittedly vague, and one may object that it excludes many phenomena that are often included in discussions of “complex predicate” constructions (e.g. resultative constructions such as *hammer the metal flat*). However, for comparative purposes, it is best to concentrate on the shared core of relevant phenomena, as I will briefly discuss in general terms in §9 below.

A *monoclausal* construction is defined as a construction that can be negated only in one way, i.e. which does not allow predicate negation in the scope of a predicate negation (see (5) above). For example, (10a) and (10b) are semantically quite similar, but only

³ Note the danger of Eurocentrism here – what is “normally” coded on verbal heads might be what we expect to be coded on the head from the perspective of European languages.

(12) French

- a. *Karim mange la soupe.*
 Karim eat the soup
 ‘Karim eats the soup.’
- b. *Fatima fait manger la soupe à Karim.*
 Fatima makes eat the soup to Karim
 ‘Fatima makes Karim eat the soup.’

In (12b), it is not possible to say that *à Karim* is an argument of *fait* while *la soupe* is an argument of *manger*. Here we clearly have a single set of three arguments of the biverbal complex predicate construction. But in many other situations, it is not obvious, because one could interpret the situation in multiple ways, e.g. in English sentences such as *she hopes to see me*. There are two overt arguments here (*she* and *me*), plus apparently an unfilled argument position (the empty subject position of *see*). And as we will see in (15) below, some monoclausal serial verb constructions have multiple object positions.

In the next three sections, I will discuss the terms *serial verb construction* (§4), *converb* (§5), and *auxiliary* (§6), which have been important in complex predicate research.

4. Serial verb construction

A serial verb construction is a subtype of biverbal predicate construction, with additional requirements, as specified in the definition in (1) that was already given in §1. Two examples are provided in (13)-(14).

(1) **serial verb construction**

A serial verb construction is a monoclausal construction consisting of multiple independent verbs with no element linking them and with no predicate–argument relation between the verbs. (Haspelmath 2016: 296)

(13) Barayin (Chadic)

Gor-e kol-e siidi.
 run-PRF go-PRF home
 ‘[He] ran home.’ (Lovestrand 2018: 83)

(14) Manus Koro (Oceanic)

You ta tay-i pu i mat pwi.
 I.SBJ NEG beat-OBJ pig 3SG.REAL die NEG
 ‘I didn’t kill the pig.’ (Cleary-Kemp 2015: 174)

The definition in (1) thus includes three crucial ingredients: (i) there is no linking element (distinguishing them from converb constructions, §5); (ii) they are independent verbs (unlike auxiliaries, §6); and (iii) there is no predicate–argument relation between the two verbs (unlike in control constructions, e.g. *she wants to sing*).

On this definition, serial verb constructions (SVCs) are biverbal constructions, but not all of them fulfill the requirements of the definition of verbal complex predicate constructions in (8), specifically the requirement of a single set of argument positions. Serial verb constructions may have two different patient arguments which are both in object position, as in (15):

- (15) Ewe (Kwa; Ameka 2006: 313)
ku tsi kló ŋkú.me
 2SG.scoop water wash face
 ‘Fetch water and wash your face.’

Let us briefly compare the definition in (1) with the characterization given by Aikhenvald (2018) in her recent monograph on serial verb constructions:

“In many languages of the world, a sequence of several verbs act together as one unit. They form one predicate, and contain no overt marker of coordination, subordination, or syntactic dependency of any other sort. Such series of verbs are known as *serial verb constructions*, or serial verbs for short. Serial verbs describe what can be conceptualized as a single event. They are often pronounced as if they were one word.” (Aikhenvald 2018: 1):

This is of course similar to the definition that I gave in 2016, but it is not quite clear whether it is intended as a definition or merely as a description of what is commonly found in languages (note the qualification “often” in the last sentence). I see four problems with Aikhenvald’s characterization: (i) it does not exclude predicate-argument constructions such as *she wants to sing*, *they made us laugh*, or maybe even *he ought to talk*, which are not normally called serial verb constructions even when they have no marker of dependency; (ii) saying that “they act together as one unit” is extremely vague; (iii) the requirement that they “can be conceptualized as a single event” cannot be operationalized and remains subjective; and (iv) there is no good general foundation for the phonological criterion of “phonological wordhood”, as mentioned in §2.1.

Of course, it is quite normal that two authors’ use of a traditional term is different, as there are no terminological committees in linguistics that would decree a certain definition and reject others. But linguists should be aware of the differences, so I will continue the comparison by reacting to Aikhenvald’s (2018: 18) criticism of my 2016 definition:

“This definition [i.e. Haspelmath’s 2016] is fairly problematic. ... The obscure formulation ‘predicate–argument relation between the verbs’ arbitrarily excludes serialization of complement-clause taking predicates and causative serial verb constructions (widespread types of serial verbs in many languages...). Sharing tense, aspect, modality, and mood are deemed to be ‘unnecessary’ criteria for serial verb constructions.”

Aikhenvald is right that my 2016 definition is (partly) “arbitrary”, like any other proposal for a general definition.⁶ However, it is motivated by the requirement of preserving traditional usage, recognizing that complement relations and causative relations are not normally treated as serial verb constructions in the literature.⁷ Including them would be a

⁶ Terms which denote natural kinds are not arbitrary, but in the present context, we are talking about comparative concepts for grammatical systems as cultural phenomena, not about natural phenomena (see §8 below).

⁷ Aikhenvald calls my term “predicate-argument relations” obscure, and it must be admitted that it is vague (I do not give a clear definition of it, relying on readers to understand it as a kind of primitive). However, what I mean is very similar to what Aikhenvald means by “complement-clause taking predicates”, so her term could be substituted in my definition. My aim was to exclude cases such as ‘she wants to sing’, ‘they made us laugh’, ‘he ought to talk’. (It has been suggested by Andrey Shluinsky, p.c., that such cases should be included in Kwa languages, but this is no problem – language-particular or family-specific terms are needed anyway. They can be included in discussions of these languages, but this has no direct bearing on the comparative-concept definition.)

terminological innovation, thus adding to the confusion. Aikhenvald's other point of criticism is actually a clear virtue of my definition: If one does not include TAM sharing in the definition, one can formulate an empirical hypothesis (see §8 below). It appears that Aikhenvald is primarily interested in characterizations that give a rich picture of the phenomenon, rather than in jointly necessary and sufficient conditions for defining a phenomenon for further hypothesis-testing.

In the context of the present volume, it should be mentioned that on the 2016 definition, verb compounds are serial verb constructions, e.g.

(16) Japanese

ジョンが鶏を殴り殺した。

John-ga niwatori-o naguri-korosi-ta.

John-NOM chicken beat-kill-PST

'John beat a chicken and killed it.' (cited by Haspelmath 2016: 298)

Such compounds are sometimes said to be "single words", but the criteria for wordhood are rarely uniform. That we cannot apply phonological criteria systematically has already been observed in §2.1. Such compound verbs are therefore included in my definition of serial verb constructions. Lovestrland (2021: n. 6) says that they are not "prototypical", but prototypes play no role in the definitions proposed in the present paper.

5. Converb

Let us next look at converbal clause constructions, whose similarities with serial verb constructions have been well-known since Bisang (1995). However, the most typical converb constructions involve subordination, as in (17)-(18).

(17) Beja (Cushitic)

i=ka:=iji haraw-a a-dif=ho:b

DEF-camel-1SG seek-CVB 1SG-leave.PFV=when

'when I left to look for my camel...' (Vanhove, this volume)

(18) Newar (Trans-Himalayan)

āmun biskut ṇar-i doṇ-an chē yer-a rā?

3SG.ERG biscuit buy-INF finish-CVB house come-3SG.PST Q

'When he finished buying the biscuits, did he come home?' (Genetti 2005: 66)

These are not biverbal constructions (or "verbal complex predicate" constructions) because are not monoclausal.

In Haspelmath (1995: 3), I defined a converb as "a nonfinite verb form whose main function is to mark adverbial subordination".⁸ However, converbs may also occur in monoclausal biverbal constructions, and this makes them relevant in the present context. For example, the construction in (19) is monoclausal and contains a converb.

⁸ The first authors to use *converb* in a general sense in typology were Nedyalkov & Nedyalkov (1987), who introduced *converb* as a term corresponding to Russian *deepričastie*. Since converbs are common in the Uralic, Turkic, Mongolic, Tungusic and Dagestanian languages of Russia and the former Soviet Union, Soviet and Russian linguists have long had a special interest in converbs.

- (19) Turkish
Ali otur-up kal-ıyor.
 Ali sit-CVB remain-PROG
 ‘Ali remains seated.’ (Kuribayashi 2021: 463)

Here I propose a new definition of *converb*, because the 1995 definition was inadequate. It contains the term “nonfinite”, which is not well-defined in a cross-linguistic perspective (see Cristofaro 2007 and Nikolaeva 2010 for some of the problems), and the part about the “main function” is vague. In addition, I did not have a good way to delimit “adverbial” subordination. The new definition is given below, repeated from (2).

(2) **converb**

A converb is a form that (i) consists of a verb stem and a special affix but no subject person indexing, (ii) that cannot occur as the predicate of an independent sentence, and (iii) that can occur as the predicate of a subordinate clause that is neither a relative clause nor a complement clause.

In this definition, reference to “adverbial” is avoided in that converbal clauses are defined negatively, as neither relative clauses nor complement clauses (i.e. adnominal modifier clauses and argument clauses, using two key notions that we saw in §3 above in the definition of biverbal predicate constructions).

While converbs cannot be defined as “nonfinite”, the definition in (2) says that the verb in a converbal clause must lack person indexing, so that subordinate clauses with person marking as in (20)-(21) are not included.

- (20) Ancash Quechua
Chakra-chaw urya-pti-i, María pallamu-rqu-n wayta-kuna-ta.
 field-LOC work-SUBOR-1SG María pick-RPST-3SG flower-PL-ACC
 ‘While I worked in the field, María picked flowers.’ (Cole 1983: 3)

- (21) Beria (Saharan; Chad)
Bàā òù-d-é kéi-d-í.
 well dig-1PL-SUBOR come-1PL-PFV
 ‘We came to dig a well.’ (Jakobi & Crass 2004: 173)

There are different uses of the term *converb* in the literature with respect to the feature of person marking. In their discussion of converbs in language of the northeastern African area, Amha & Dimmendaal (2006) discuss the “finiteness” issue, and they opt for the use of the *converb* term in a very broad sense (following V. Nedjalkov 1995, who did not exclude person-marked verb forms either). But since the most typical converbs (of the Turkic and Slavic type, as well as Romance gerunds) lack person marking, it is better to exclude person-marked forms, because otherwise many other kinds of verb forms would have to be included (perhaps even some Indo-European forms that would normally be called “subjunctives”).

Converbs typically have their unique affix, though the verb in a converbal clause may also have an affix that also occurs elsewhere, as in English *-ing* clauses, which can also occur as relative clauses (I. Nedjalkov 1998 calls such converbs “non-strict”). The definition in (2) specifies that converbal forms should be marked by an affix, and this is indeed the case with the great majority of forms that have been called “converbs”.⁹

⁹ However, I. Nedjalkov (1998) mentions “converbs” that are marked suprasegmentally or that have the shape of the simple verb stem. Such forms are not included here, because with suprasegmental marking, it

Converbal clauses must be subordinate, which means that “narrative converbs” (V. Nedjalkov 1995) are not included here, and neither are clause-chaining constructions in which each clause is asserted. Subordinate clauses are characterized by positional mobility, or internal position, or focusability, or a combination of these (Haspelmath 1995: 12-17). For example, Cantonese converb forms ending in *-can* can occur not only in peripheral position as in (22a), but also in internal position as in (22b).

(22) Cantonese (Yip 2022: 144; 149)

- a. *Go bibi siu-can, go hau zau maak-dou hou daai.*
 CLF baby laugh-CVB CLF mouth then open-till very big
 ‘Whenever the baby laughs, his mouth opens wide.’
- b. *Ngo sik nigo lai-can zau bei jan naau ge hoksaang.*
 1SG know this [come-CVB then PASS person scold REL] student
 ‘I know this student who got scolded whenever he came.’

Similarly, Japanese converb forms ending in *-te* can occur in multiple positions, showing that it qualifies as subordinate (Nakatani 2016: 401-402):

- (23) a. 母が財布を持って逃げた
Haha-ga [saihu-o mot-te] nige-ta.
 mother-NOM wallet-ACC hold-CVB run.away-PST
 ‘The mother ran away holding her wallet.’
- b. 財布を持って母が逃げた
[Saihu-o mot-te] haha-ga nige-ta.
 wallet-ACC hold-CVB mother-NOM run.away-PST
 ‘The mother ran away holding her wallet.’
- (24) a. 父が帰って娘は喜んだ。
[Titi-ga kaet-te] musume-wa yorokon-da.
 father-NOM return-CVB daughter-TOP get.delighted-PST
 ‘The father came back, and the daughter got delighted.’
- b. 娘は父が帰って喜んだ。
Musume-wa [titi-ga kaet-te] yorokon-da.
 daughter-TOP father-NOM return-CVB get.delighted-PST
 ‘The father came back, and the daughter got delighted.’

Contrasting with this, there are many languages that have verbal affixes which can only be used in dependent clauses which are not subordinate (or embedded) in this sense. These are included as “converbs” in the broad definition of V. Nedjalkov (1995), but they are not included here. For example, the “same-subject sequential” forms of Mauwake, which are marked by *-op/-ep*, are not subordinate because they always occur in a peripheral position, e.g.

(25) Mauwake (Trans-New Guinea; Berghäll 2015: 152)

Owowa or-op, wuailal-ep akia ik-e-k.

is not clear how the various forms relate to each other, and with “zero-marking”, there is an even greater risk that the marking is not a fact of the language, but merely an artifact of the analysis.

village go-SS.SEQ be.hungry-SS.SEQ banana roast-PST-3SG
 ‘He went to the village, became hungry, and roasted bananas.’

It must be noted that in many languages, one and the same form can be used in a subordinate converbal clause as defined in (2) and also in a clause-chaining construction. These forms are still regarded as converbs, just as participial clauses such as English *-ing* clauses are converbs even though they also occur in relative clauses. For example, Turkish *-up/-ip* converbs can be used both in adverbial clauses as in (26a) and in dependent (but nonsubordinate) sequential clauses such as (26b).

(26) Turkish (Johanson 1995: 333)

a. *Otur-up mu konuş-tu-lar?*
 sit-CVB Q speak-PST-3PL
 ‘Did they speak while they were sitting?’

b. *Oda-ya gir-ip otur-du-m.*
 room-DAT enter-CVB sit-PST-1SG
 ‘I went into the room and sat down.’

Japanese *-te* converbs seem to be similar because they can also be used in purely sequential situations. Sentence (27) is ambiguous between two different meanings, a sequential and a non-sequential (subordinate) one (as discussed by Oshima 2012: 305).

(27) *Hiroshi-wa tat-te shashin-o tot-ta.*
 Hiroshi-TOP stand-CVB photo-ACC take-PST
 ‘Hiroshi stood up and took a photo.’
 OR: ‘Hiroshi took a photo standing on his feet.’

In many languages, some converbal forms occur in monoclausal constructions, not only in subordinate converbal clauses. Two further examples are given in (28) and (29) (see also (19) above).

(28) Japanese (Shibatani 2009: 257-258)

Kare=ga koko=ni hon=o mot-te ki-ta.
 he=NOM here=DAT book=ACC take-CVB come-PST
 ‘He brought the book here.’

(29) Marathi (Pardeshi 2021: 231)

Tyā mulālā-ne ma-lā kholi sāp kar-un dili.
 that boy-ERG me-DAT room clean do-CVB gave
 ‘That boy cleaned the room for me.’

Since my 1995 paper, several authors have discussed the definition of the term *converb* (van der Auwera 1998; Ylikoski 2003; Coupe 2006; Rapold 2007), which reflects the fact that the 1995 definition was not clear enough. There is no space to discuss these earlier papers here, but I hope that it has become clear how the current definition of *converb* conforms to the criteria for a good definition that we saw in §2.

6. Auxiliary

Auxiliary verbs have not often been treated under the “complex predicate” heading (but see Anderson 2011; Godard & Samvelian 2021). I discuss the term *auxiliary* here because some kinds of elements that are usually regarded as auxiliaries have been included in the “verbal complex predicate” class.

For a long time, it was quite unclear to me how *auxiliary* could be defined, and I still do not know a definition that would be fully satisfactory. However, *auxiliary* is such an important term that it needs to be retro-defined (as per the requirement in 4c above). In Anderson’s cross-linguistically oriented monograph, an auxiliary verb is said to be

“an item on the *lexical verb–functional affix* continuum, which tends to be at least somewhat semantically bleached, and grammaticalized to express one or more of a range of salient verbal categories, most typically aspectual and modal categories, but also not infrequently temporal, negative polarity, or voice categories.” (Anderson 2006: 4-5)

This characterization is a good description of the problem: We all know that (“full”) verbs often develop diachronically into forms that indicate “grammatical” meanings, but if this process is gradual, it should give rise to a synchronic “continuum”. Defining *auxiliary* in such a way that the cut is made along some arbitrary point on the continuum would not seem to be helpful for understanding. However, the continuum is generally thought to be a multidimensional continuum, and it is an empirical question to what extent the different dimensions correlate with each other (see Bisang et al. 2020 for some initial quantification). This makes it difficult to apply the grammaticalization criterion to the definition of *auxiliary*.

A recently proposed alternative is Boye’s (2023a) approach to grammaticalization, which rejects the continuum view and makes a clear-cut distinction between lexemic and grammatical forms. His distinction between discursively primary and secondary forms is very similar to the distinction between *free* and *bound* forms which I used in other papers to define the terms *affix* and *clitic* (Haspelmath 2021a). Thus, I propose to cut the Gordian knot and adopt a comparative-concept definition of *auxiliary* in terms of the notion of a bound form that is not an affix, as in (3) (repeated here from §1 above).

(3) auxiliary

An auxiliary is a non-affixed bound form cooccurring with a verb that expresses TAMEP (tense, aspect, modality, evidentiality, or polarity) meanings and that has person marking which indexes the verb’s subject.

Like all definitions of comparative concepts, this definition makes no reference to categories or classes, but only to purely formal concepts (“bound”, “non-affixed”) and to purely functional-semantic concepts (tense, modality, etc.).

An auxiliary is not defined as a kind of verb here, because *verb* is defined semantically (as a root denoting an action, Haspelmath 2023d: 34). Instead, *auxiliary* is defined semantically as well, as a clitic form expressing TAMEP meanings.¹⁰ The particular choice of meanings is of course arbitrary, and one could have made different choices. The reason I chose TAMEP (tense, aspect, modality, evidentiality, or polarity) in (3) is that these are the most common kinds of meanings mentioned for auxiliaries. A criterion that is often mentioned is that auxiliaries do not assign additional semantic roles, so that forms

¹⁰ A bound morph that is neither an affix nor a root (a noun, a verb or an adjective) is a *clitic* (Haspelmath 2023c). Thus, auxiliaries are clitics.

meaning ‘want’, ‘try’, or ‘cause’ are excluded for semantic reasons. The TAMEP categories are also the kinds of meanings that are normally treated as verbal inflectional meanings (Haspelmath 2024b), and auxiliaries are often thought of as expressing the same kinds of meanings that can be expressed by inflectional affixes.¹¹

But not every non-affixed bound form expressing these kinds of meanings is considered an auxiliary. For example, negator particles such as Italian *non* ‘not’ express polarity but are not auxiliaries, and the same is true of evidential particles, as illustrated by the reportative particle *paá* in (30).

- (30) Nheengatu (Aikhenvald 2015: 244)
U-sú u-piniatika paá.
 3SG-go 3SG-fish REPORTATIVE
 ‘He went fishing (they say/I was told).’

In practice, the literature often sees auxiliaries as elements that derive from verbs but have acquired grammatical meanings, and when the diachronic origins are known, this is of course a valid perspective (e.g. Heine 1993). But for a synchronic definition, we need a synchronic criterion, and this is offered by subject person marking. Thus, the Finnish negating form *e-* in (31a) and the Maltese phasal-polarity form *ghad-* in (31b) count as auxiliaries, while modals such as Mandarin *néng* ‘can’ in (32) do not count as auxiliaries.¹²

- (31) a. Finnish
E-mme leiki puutarha-ssa.
 NEG-1PL play garden-INESS
 ‘We are not playing in the garden.’
 b. Maltese
Jien għad-ni n-iktieb.
 I write-1SG 1-write.IMPFV.SG
 ‘I am still writing.’ (Camilleri 2016: 176)
- (32) Mandarin Chinese
 他不能騎馬
Tā bù néng qí mǎ.
 he not can ride horse
 ‘He cannot ride a horse.’ (Tsang 1981: 39)

That subject person marking must be present is of course an arbitrary component of the definition in (3). Anderson (2006: Chapter 3) explicitly allows for auxiliaries that do not show person marking and that combine with person-marked main verbs (“LEX-headed auxiliaries”). Examples come from Modern Greek and from Yucatec Maya:

¹¹ Grammatical voice is omitted from this list because voice affixes are very often treated as derivational, and because voice markers are not neutral with respect to semantic roles. (In Haspelmath 2024b, I excluded aspect from inflectional meanings because aspectual affixes are very often treated as derivational. These decisions are somewhat arbitrary, and should not be taken as more than terminological.)

¹² More generally, languages that lack subject person marking on verbs cannot have auxiliaries according to the definition. This is unintuitive, but I do not see an alternative.

- (33) a. Modern Greek
Prépi na ðulép-s-is.
 must SBJV work-PFV-2SG
 ‘You have to work.’
- b. Yucatec Maya
Bíin líub-uk-en.
 FUT fall-SBJV-1SG
 ‘I will fall.’ (Lehmann 2017: 192)

While the Modern Greek form *prépi* ‘must’ is definitely verb-like in that it has a regular past tense form with prefix *é-* (*é-prep-e na ðulép-s-is* ‘you had to work’), the same is not true for the Yucatec Maya future marker *bíin*. Nevertheless, Lehmann (2017: 231) wants to treat it as an auxiliary, probably because it originates in an earlier verbal form. However, definitions for comparative purposes must be strictly synchronic (cf. §2.1 above), so the Yucatec “auxiliaries” cannot be included. The Greek form for ‘must’ in (33a) can be said to include a person suffix (*prép-i* [necessary-3SG]), but it is not the person of the subject of the cooccurring verb, so it does not fulfill the requirement of the definition. Of course, the choice of subject person marking, rather than, say, tense, is somewhat arbitrary, but it seems to me that of all the “finiteness” features, subject person marking is the feature that is most often associated with typical auxiliaries in the literature.

An important aspect of the definition in (3) is that an auxiliary is defined as a *bound* form cooccurring with a verb. This means that when a form can occur on its own without a verb, it is not an auxiliary. Consider the two Spanish sentences in (34a-b). With the infinitive-taking modal verb *querer* ‘want’, the main verb ‘play’ can be omitted when it is given in the context, as seen in (34a). But with the tense auxiliary *había* in (34b), this is not possible: *Había* is a bound form that must always cooccur with a verb. *Querer* ‘want’, by contrast, is not an auxiliary by this criterion.

- (34) a. *Pregunté a María si quería jugar. La niña quería.*
 I.asked ACC María if she.wanted to.play the girl wanted
 ‘I asked María if she wanted to play. The girl wanted to.’
- b. *Pregunté a María si había jugado. *La niña había.*
 I.asked ACC María if she.had played the girl wanted
 ‘I asked María if she had played. The girl had.’

This criterion gives unintuitive results for English, where *have/has/had* can occur without the main verb (as in the translation of (34b)), but it seems to be quite rare for tense-aspect markers and related forms to be able to occur without the main verb. The reason for including the “bound form” criterion in the definition is that this is the main criterion for grammaticalized status, as briefly discussed in the next section.

One might ask what term should be used for such phenomena if they are not auxiliaries. No well-known term comes readily to mind, but we should not expect traditional terms to cover the phenomena of the world’s languages exhaustively. Perhaps we will ultimately need a rather different typology with different terms, but this is not the topic of the present paper.

7. Grammaticalization and inventorization (or “lexicalization”)

Now that we have seen which issues arise in defining serial verb constructions, converbs and auxiliaries, let us step back and consider the conceptual issues arising with grammaticalization and “lexicalization”, two diachronic concepts that are important in complex predicate research.

As is well-known, constructions with two full verbal forms that are closely linked semantically often give rise to “analytic constructions”, or to fixed expressions, or to expressions that are both analytic and fixed. Typical examples are given in (35)-(36).

(35) a typical analytic expression: Croatian

Ne-će-š pjevati.
 NEG-FUT-2SG sing
 ‘You will not sing.’ (-će- ‘FUT’ > *htjeti* ‘want’)

(36) a typical fixed expression: German

fallen lassen
 fall let
 ‘abandon sth.’ (lit. let sth. fall)

An “analytic construction” is defined as including an open slot for a lexeme and a grammatical marker, and a fixed expression is defined as having an unpredictable meaning or otherwise unpredictable properties. These kinds of expressions are generally thought to arise via grammaticalization and “lexicalization”, respectively.

However, these two diachronic processes lead away from biverbal predicate constructions as defined in §3: If a verb is grammaticalized to become an auxiliary (as defined in §6), it is no longer a verb in the sense of the original definition, because it acquires a meaning that is no longer an action meaning. And if a combination consisting of two verbs acquires an idiomatic meaning, it no longer instantiates a construction, because constructions are by definition semantically regular (I discussed this for serial verb constructions in Haspelmath 2016: §2.1).

Thus, biverbal predicate constructions arise from multiclausal constructions by a process of monoclausalization, not necessarily by a process of grammaticalization, and never by a process of “lexicalization”. The neologism *monoclausalization* is defined as in (37). (This is similar to, but broader than, what Harris & Campbell (1995: §7.4) called *clause fusion*.)

(37) **monoclausalization**

Monoclausalization is the change by which a previously biclausal construction becomes monoclausal.

But how exactly are the terms “grammaticalization” and “lexicalization” defined? Just as synchronic grammatical terms often have different meanings, terms for diachronic processes or types of changes are sometimes unclear, and the definition of “grammaticalization”, in particular, has given rise to some discussion (e.g. Haspelmath 2004; von Mengden & Simon 2014). Here I will make concrete proposals for a retro-definition of *grammaticalization*, and instead of defining “lexicalization” (a term that has been used in multiple incompatible meanings), I offer a definition of the related term *inventorization*.

As noted briefly in the preceding section, Boye (2023a; 2023b) defines a grammatical element as an element that is by convention discursively secondary. Grammaticalization

can then be regarded as a change by which an expression that originally had a potentially primary role in discourse becomes a discursively secondary expression. An expression that is potentially discursively primary is more or less the same as an expression that is free (in Bloomfield's 1933 terms), and an expression that is by convention discursively secondary is more or less the same as an element that is bound and does not denote an action, an object or a property. Thus, grammaticalization can simply be defined as the rise of a grammatical marker from a free form, as in (38).

(38) **grammaticalization**

Grammaticalization is the change from a free form to a grammatical marker.

This definition is less sophisticated than Boye's, but it amounts to more or less the same. Well-known kinds of grammaticalization changes that are subsumed by this definition are the change types in (39).

(39)	noun	>	adposition
	verb	>	auxiliary
	demonstrative	>	determiner
	numeral	>	determiner
	interrogative pronoun	>	relativizer
	verb	>	subordinator (e.g. <i>suppose</i>)

It does not include changes such as “adposition > case affix”, or “auxiliary > TAM affix”, but the evidence for such changes is surprisingly weak (Haspelmath 2022).

The definition in (38) is very similar to Meillet's (1912) original characterization of grammaticalization (“l'attribution du caractère grammatical à un mot jadis autonome”),¹³ and I now think that it is better than the definition that I suggested two decades ago (“a grammaticalization change is a diachronic change by which the parts of a constructional schema come to have stronger internal dependencies”; Haspelmath 2004: 26). At that time, I thought that grammaticalization must be seen as a kind of gradual change, because there seem to be clear “degrees of grammaticalization”. However, I have come to realize that if we wanted to quantify such degrees, we would need ways of measuring them, and this is still very difficult (though Bisang et al. 2020 have made an important start). In any event, we need some basic concepts that are not gradient, because otherwise we cannot demonstrate the gradience.

Thus, I now prefer a discrete or categorical (i.e. non-gradient) distinction between grammatical markers and free forms, and I treat this as the only defining criterion of grammatical status, largely following Boye (2023b). The other features that are typically present in grammaticalization (“semantic bleaching”, “phonological erosion”, positional fixation, etc.) can be seen as following from the definition, or as secondary developments. I will not elaborate on this here, but in the context of biverbal predicate constructions, it should be noted that the development of an auxiliary from a complement-taking verb (or the development of an adposition from a serial verb) leads us away from biverbal constructions. Auxiliary constructions such as Croatian *ne-će-š pjevati* ‘you will not sing’ (in 35) are not biverbal, and they would fall under the definition of “verbal complex predicate” only if the future auxiliary *će-* were treated as somehow still being the same verb as *htjeti* ‘want’, from which it originates.

Next, let us consider “lexicalization”, by which linguists often mean a change by which a composite expression that is constructed in a fully predictable way comes to have

¹³ “the attribution of grammatical character to a word that was formerly autonomous”

idiosyncratic properties so that it must be “listed in the lexicon”. Many languages have unpredictable combinations of serial verbs and of converbs with independent verbs, e.g. Sranan *bro kiri* [blow kill] ‘extinguish’ (Durie 1997: §3.2). Linguists often say that such combinations are “lexicalized”. (Another example is German *fallen lassen* (lit. ‘let fall’) ‘abandon’, which we saw in (36) above.)

But the terms “lexicon” and “lexical” have several different meanings, so a more precise definition of the process refers to the *inventorium*, i.e. the set of forms and constructions that cannot be predicted from other forms or constructions (Haspelmath 2024a). The *inventorium* is different from the mental lexicon because the latter also contains many composite expressions that are fully predictable, and also from the “lexemicon”, i.e. the set of all lexemes of a language (which is often contrasted with the “syntactic component” of grammar). We can thus define *inventorization* as in (40).

(40) inventorization

Inventorization is a diachronic change by which a composite expression acquires unpredictable properties so that it must be part of the *inventorium*.

Thus, inventorization is not merely about holistic mental storage by individual speakers, but about the diachronic development of conventional idiosyncrasies across the community. And it is not (necessarily) about a shift from a syntactically composed expression to a “word”-like expression (this would be *univerbation*, as recently discussed by Lehmann 2020; or maybe “lexemization”, when a new lexeme arises).

It appears that inventorization is particularly common with biverbal predicate constructions, perhaps for the same reason that nominal compounding often involves inventorized (and idiomatized) meanings. However, it is not clear why this should be so, and this impression has not been quantified systematically. Clear and simple definitions will help future research to gain more insights here.

8. Definitions versus hypotheses

The bulk of this paper has focused on definitions, because we need to be clear about our terms before we can formulate and test hypotheses.

In addition, we need to be clear about the level at which our hypotheses are formulated. In research traditions such as HPSG (Godard & Samvelian 2021) and LFG (Andrews 2023), what matters primarily is elegant and complete language-particular analyses (what I have called “p-linguistics”, Haspelmath 2021b). In such contexts, the hypotheses are language-particular, and the terminology and concepts can be different in different languages, depending on the particular needs.¹⁴

In other research traditions, the concepts are applied cross-linguistically and have relevance for general theories (“g-linguistics”): They may either be hypothesized to be elements or categories of an innate “universal grammar” blueprint (this was called the “categorical universalist” view in Haspelmath 2010), or they may be thought of as comparative concepts which are employed as methodological tools. On the UG-oriented view, concepts such as “auxiliary” (cf. AUX in Chomsky 1965) or “clause” (cf. S, or CP, in mainstream generative grammar) could be thought of as innate categories, and thus universal hypotheses. If they are innate and thus natural kinds, one could use different

¹⁴ In these research traditions, there is often some awareness of the possibility that different-looking analyses may in fact just be notational variants of each other (see also Dryer 2006). For example, Przepiórkowski (2023: 1870-1871) compares LFG and HPSG and for illustration presents concrete “LFG-like” HPSG representations as well as “HPSG-like” LFG representations of example constructions.

criteria for them in different languages. This has often been done in the generative literature, also in some of the HPSG and LFG literature. For example, Butt (2014: 171) is quite explicit about this in her characterization of complex predicates:

“Complex predicates are formed when two or more predication elements enter into a relationship of co-predication. ... Tests for complex predicatehood are language specific.”

In this research tradition, general terms can be tested both by applying them to language-particular analysis and to cross-linguistic generalizations.

If, by contrast, such categories are not claimed to be innate, the cross-linguistically applied concepts are comparative concepts, and must be defined uniformly, as noted in §2.1 above. As comparative concepts, they are not defined structurally but in terms of semantic or formal substance, and they cannot be used to describe particular languages (i.e. to express hypotheses about particular languages). They are ideally suited, however, for expressing cross-linguistic hypotheses or claims.

As a concrete illustration of the distinction between definitions and hypotheses, let us now briefly consider two hypotheses about biverbal predicate constructions that have some initial plausibility. The first is adapted from Baker & Harvey (2010), who propose a hypothesis along these lines for “complex predicates”. The second is advanced tentatively here, on the basis of general considerations and impressionistic evidence.

- (41) **The biverbal predicate causal chain hypothesis** (cf. Baker & Harvey 2010: 21)
Biverbal predicate constructions describe complex situations in which the meaning components ‘cause’, ‘become’ and ‘be’ occur only once and in this order if they occur.
- (42) **The biverbal predicate frequency hypothesis**
Biverbal predicate constructions describe complex situations which are particularly frequently talked about.

As Krauß & Harvey (2021: 103) observe, it is remarkable that the causal chain of (41) applies in the same way to single-morph predicates and to “complex predicates”. Typical examples of biverbal predicates obeying the causal chain are ‘beat...die’ in (14) above, and ‘take...come’ in (28) above. According to this hypothesis, complex situations of the type ‘become...cause’ are not expressed by biverbal predicate constructions. By the same token, single-morph verbs never express such “inverted” causal situations.

It appears that the generalization about the semantics of biverbal predicate constructions correlates with frequency of use: We talk significantly more often about situations in which ‘cause’ precedes ‘become’, which precedes ‘be’ in the event chain. This can be tested by examining corpora of languages that lack biverbal predicate constructions and that therefore must convey these meanings in different ways. The expectation is that we much more often find sequences like ‘I knocked the vase over, it broke and was worthless’ than sequences like ‘the vase was old, it fell and hit me’. The conventionalization of specific patterns as biverbal predicate constructions may well be a simple consequence of the higher frequency of certain event sequences in discourse. Likewise, the higher frequency of ‘cause’ > ‘become’ > ‘be’ sequences leads to the frequent expression of such meanings by single-morph verbs (e.g. ‘kill’, which basically means ‘cause (become (dead))’, and which is often expressed by a single morph).

Presenting evidence for these two hypotheses (or against them) is outside the scope of the present paper and is left as a task for future research. The purpose of citing these hypotheses is to make clear why we need clear definitions for larger questions. Ambitious

general hypotheses and general theories cannot be formulated unless we have clearly defined comparative concepts.

9. Concluding remarks

In this paper, the focus has been on providing clear and straightforward retro-definitions of the three terms *serial verb construction*, *converb*, and *auxiliary*. Linguists use these terms regularly, and they expect to be understood by other linguists, so it should be useful to have such definitions, as well as some background discussion.

This paper is a methodological contribution, and it has no immediate consequences for answering specific research questions. It needs to be emphasized that agreement on definitions of grammatical terms is not normally an outcome of research. But again and again, one finds comments in the literature suggesting that as more research is done, one expects more consensus on the definitions of terms such as “auxiliary”, “multi-verb construction” or “light verb construction”, e.g.

Nolan & Diedrichsen (2017: 2)

“Many different definitions of multi-verb constructions exist in the literature and it is clear that there is **(still) no unified consensus** on the characteristics of these in the world’s languages.”

Fleischhauer & Hartmann (2021: 136)

“Although light verb constructions (LVCs) have been analyzed from many different perspectives for decades, **there still exists no consensus** regarding some fundamental issues. Even the very definition of the concept is still subject to debate.”

In fact, the opposite is often the case: The more widely a grammatical term is used, the more different meanings it has, and the less agreement there is about the nature of the relevant phenomena. While more research means more knowledge about languages, it does not necessarily translate into additional general insights, because it may not be clear how to compare the languages in such a way that generalizations can be formulated.¹⁵

I do not claim that the definitions offered here will be ideal for finding generalizations and deeper insights into language structures, but they are clearer than all other definitions that I know of (including my own 1995 definition of *converb*). Thus, I hope that they will help other linguists in their thinking about the issues involved in the corresponding phenomena.

The definitions in (1)-(3) are retro-definitions in the sense that they define terms that are already present in the literature, and they are formulated in such a way that the great majority of earlier uses fall under the definitions. The choices I made are sometimes arbitrary, but this is a necessary feature of comparative-concept definitions: While natural-kind terms refer to entities that are part of nature, comparative concepts are methodological tools devised by scientists, and they are always based on partially arbitrary decisions. And as noted at the end of §6, I do not suggest that the concepts discussed here are exhaustive in any way. I was asked by a colleague why “passive auxiliaries” are not included in §6, and why “converbs with subject person indexes” were not included in §5 (Andrey Shluinsky, p.c.). The reason is that the definitions would get much more cumbersome, and such phenomena are not very common. It must always be

¹⁵ Thus, while Coupe (2006) is probably right when he says that “typological research has tried to establish whether [terms like *converb*], born of their different descriptive traditions, actually represent a single grammatical category”, but as there are no clear criteria for what is a “single grammatical category”, this does not seem to be a good research question.

kept in mind that language-particular descriptions are free to define their own terms, which are rarely fully transparent anyway (because different languages have different categories). There is no intention in this paper to set up a list of terms that provide an exhaustive typology of the attested phenomena, which would go far beyond its scope.

In addition to the retro-definitions for the four terms in the subtitle of this paper, I also defined three novel terms: *biverbal predicate construction* (introduced in the present paper), *inventorization* (introduced in Haspelmath 2024a), and *monoclausalization* (used here for the first time, in (37) above). Moreover, in practice comparative linguists also make use of *ad hoc definitions*, a third type of comparative-concept definition, much of the time. Very often, the domain of investigation needs to be narrowed in a specific way in a specific project, so a comparative concept must be defined in a particular narrow way. For example, Bybee & Easterday (2019) study consonant strengthening, which is defined as “an increase in the magnitude or duration of an articulatory movement” (2019: 271). However, in the specific study that they report on, they restrict their attention to the “relative degree of oral vocal tract constriction involved in producing a consonant”. This is a narrower comparative concept, specific to their particular 2019 study, and not intended as a definition of the more general term *consonant strengthening* – in other words, it is an *ad hoc* comparative concept.

Finally, it should be noted that most retro-definitions of well-known terms are shared-core definitions, which means that they are intended only to cover only the core of the relevant phenomena in the languages that are compared. For example, in many languages with serial verb constructions, not only verbs (i.e. action-denoting roots) can occur in such constructions, but adjectives (i.e. property-denoting roots) as well. For example, François (2004) discusses serial verb constructions in Mwotlap, and cites not only examples like *hō mōl* [paddle return] ‘paddle back’, but also *bōl madamdaw* [hammer soft] ‘soften by hammering’ (2004: 111-114). The latter is not a serial verb constructions according to the definition in (1), but it is of course the same construction in Mwotlap. Similarly, languages may have converb-like forms that are marked suprasegmentally and thus do not fall under the definition in (2) (see n. 9), but they would still be subsumed under the same language-particular category. Language-particular constructions rarely map fully onto general concepts intended for comparative purposes. They are defined by language-particular structural criteria, so they typically carve out a somewhat different range of phenomena. What matters for successful comparison across languages is that the retro-defined comparative concepts cover a core of phenomena shared across languages.

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